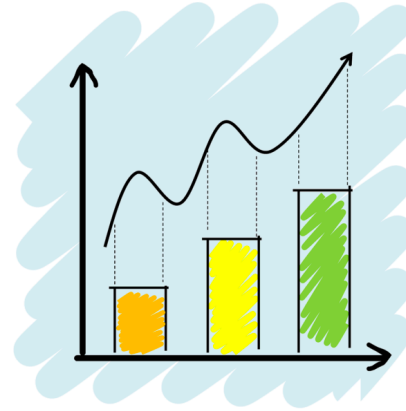


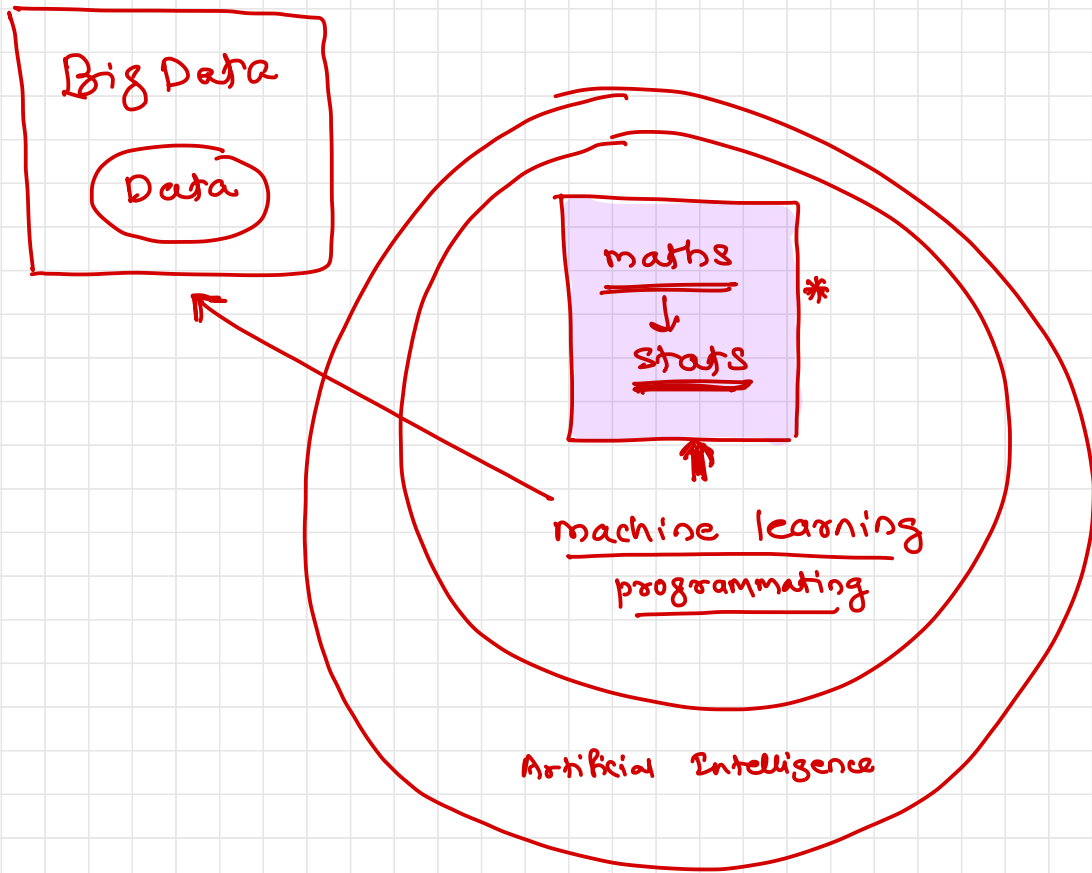


Statistics



Statistics

- ✓ - introduction
- ✓ - population & sample
- ✓ - types
 - measures of
 - ✓ central tendency
 - ✓ dispersion
 - skewness
 - kurtosis
- ✓ data & types of data
 - distributions
 - permutation & combination
 - probability
 - probability distributions
 - hypothesis testing
 - central limit theorem
 - time series analysis



Introduction

- Discipline that concerns the collection, organization, displaying, analysis, interpretation and presentation of data
- A branch of mathematics dealing with the collection, analysis, interpretation, and presentation of masses of numerical data
- Statistics has two meanings, as in plural sense and in singular sense
 - In plural sense, it means a systematic collection of numerical facts - data
 - In singular sense, it is the science of collecting, classifying and using statistics
- Is both science of uncertainty and technology of extracting information from data
- Is used help us to make the decisions
 - ↳ tools, algorithms, methods



Stages of investigation

✓ Collection of data

- It is the first stage of investigation and is regarding collection of data
- It is determined that which method of collection is needed in this problem and then data are collected

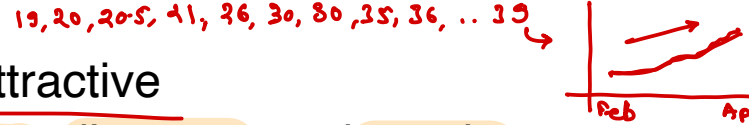
✓ Organization of data

- The data are simplified and made comparative and are classified according to time and place

→ survey → digital, manual, physical, poll., purchase data

✓ Presentation of data

- Organised data is made simple and attractive
- The data presented in the form of tables, diagrams and graphs



① storing
↳ databases
↳ apps

② data cleansing
↳ fix data type
↳ missing data
↳ removing noise

✓ Analysis of data

- To get correct results, analysis is necessary
- It is often undertaken using Measures of central tendencies, Measures of dispersion, correlation, regression and interpolation etc

✓ Interpretation of data

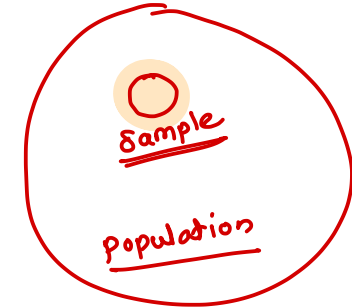
- In this last stage, conclusions are enacted
- Use of comparisons is made
- On this basis, forecasting is made



Population and Sample

- Population

- A group that has been designated for gathering the data from
- Population is always defined first, before starting the data collection process for any statistical study
- Every member is considered
- It is impossible to contact every member if the population is very large
- Represented by **N**



- Sample

- It is the part of population which is selected randomly for the study
- Subset or small portion of population
- The sample should be selected such that it represents all the characteristics of the population
- The process of selecting the subset from the population is called **sampling** and the subset selected is called the **sample**
- Represented by **n**

Sampling

- Process of getting a subset out of the population
- One of the great benefits of sampling is that a reasonably sized random sample will almost always reflect the population
- What is the challenge then ?
 - How to avoid bias?
 - How to choose sample?



Sampling Bias

- A bias in which samples are collected in such a way that some members of the intended population have a lower sampling probability than others
- It results in a **biased sample**, a non-random sample of a population (or non-human factors) in which all individuals, or instances, were not equally likely to have been selected
- If this is not accounted for, results can be erroneously attributed to the phenomenon under study rather than to the method of sampling
- Types
 - Selection Bias
 - Survivorship Bias



Selection Bias

- Favors those members of a population who are more inclined and able to answer the questions
- Types
 - ① • Under-coverage Bias
 - Making too few observations or omitting entire segment of a population
 - E.g. hospital survey of employee conducted during daytime hours. Neglects the employees working in night shift
 - ② • Self-selection Bias
 - People who volunteer may differ significantly from those in the population who don't
 - E.g. an online survey about sports team. Only people who feel strongly about the team will answer the survey
 - ③ • Healthy-user Bias
 - The sample may come from a healthier segment of the overall population – people who walk/jog, work outside, follow healthier behaviours, etc.
 - E.g. Polling customers at a fruit stand to study a connection between diet and health. Those polled likely do other things that have greater impact on their health



Survivorship Bias

- If a population improves over time, it may be due to lesser members leaving the population due to death, expulsion, relocation, etc.



Sampling methods

- Describes how the sampling can be carried out

- Types

- Radom sampling
- Stratified Random sampling
- Cluster sampling

- these are not for minimizing/removing bias



Radom sampling ←

- Random sampling means every member of a population has an equal chance of being selected
- However, since samples are usually much smaller than populations, there's a chance that entire demographics might be missed



Stratified random sampling

- Stratified random sampling ensures that groups within a population are adequately represented
- First, divide the population into segments based on some characteristic ← should be selected based on the data & experiment
- Members cannot belong to two groups at once [mutually Exclusive] → height
→ age
→ weight
- Next, take random samples from each group
- The size of each sample is based on the size of the group relative to the population
- E.g.
 - A company wants to conduct a survey of customer satisfaction
 - They can only survey 10% of their customers
 - They want to ensure that every age group is fairly represented

20-29	30-39	40-49	50-59	Total
1400	4450	3200	950	10000

- To obtain a 10% sample, take 10% from each group

20-29	30-39	40-49	50-59	Total
140	445	320	95	1000



Cluster sampling

- The idea is to break the population down into groups and sample a random selection of groups, or clusters.
- Usually this is done to reduce costs.
- E.g. Surveying only handful of neighborhoods

cluster → members closer to each other



Parameter Vs Statistics

<u>Parameter</u>	<u>Statistics</u>
<u>Is a measure that describes entire population</u>	<u>Is a measure that describes only a sample of population</u>
E.g. - <u>Actual voter turnout</u> - <u>Age of every person in country</u>	E.g. - <u>The portion of people in opinion polls</u> - <u>Age of people in sample</u>

↓
Population

↓
Sample



Statistics types

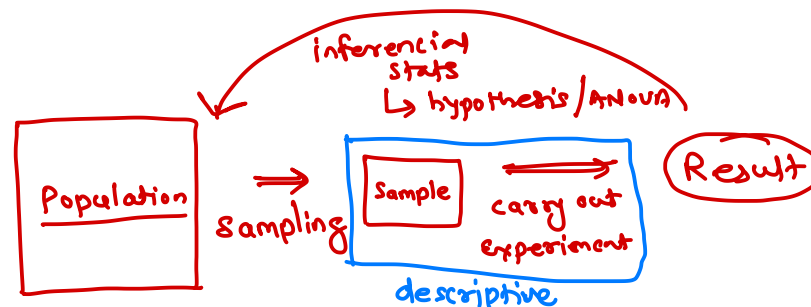
• Descriptive Statistics

- It describes the important characteristics/ properties of the data using the measures the central tendency like mean/ median/mode and the measures of dispersion like range, standard deviation, variance etc
- Data can be summarized and represented in an accurate way using charts, tables and graphs

sample → 1000 → | ch1 | ch2 | ch3 |

• Inferential Statistics

- It is about using data from sample and then making inferences about the larger population from which the sample is drawn
- The goal of the inferential statistics is to draw conclusions from a sample and generalize them to the population
- It determines the probability of the characteristics of the sample using probability theory
- The most common methodologies used are hypothesis tests, ANOVA etc

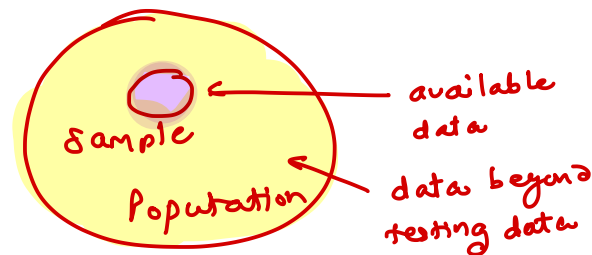


ANOVA



Statistics types

Descriptive	Inferential
Concerned with describing the target population sample	Make inferences from sample and generalize them for the entire population
Organize, analyze and present the data in meaningful manner	Compare, test and predict the future outcomes
Final results are shown in the form of charts, tables or graphs	Final result is probability score
Describes the data already known [past data]	Tries to make the conclusion about the population that is beyond the data available
Tools: - Measures of central tendency - Measures of dispersion	Tools - Hypothesis testing - Analysis of variance



Individuals and Variables

- Meaning outside the statistics
- Individuals
 - Are the people
 - E.g. 60 students for the course
- Variable
 - Is the factor that can vary
 - E.g. time the shop can make cake

- Meaning inside the statistics
- Individuals
 - Objects included in the study
 - E.g. records, people, objects
- Variable
 - Characteristics of individuals to be measured or observed
 - E.g. age or name of the person

individual → row
variable → column



Individuals and Variables

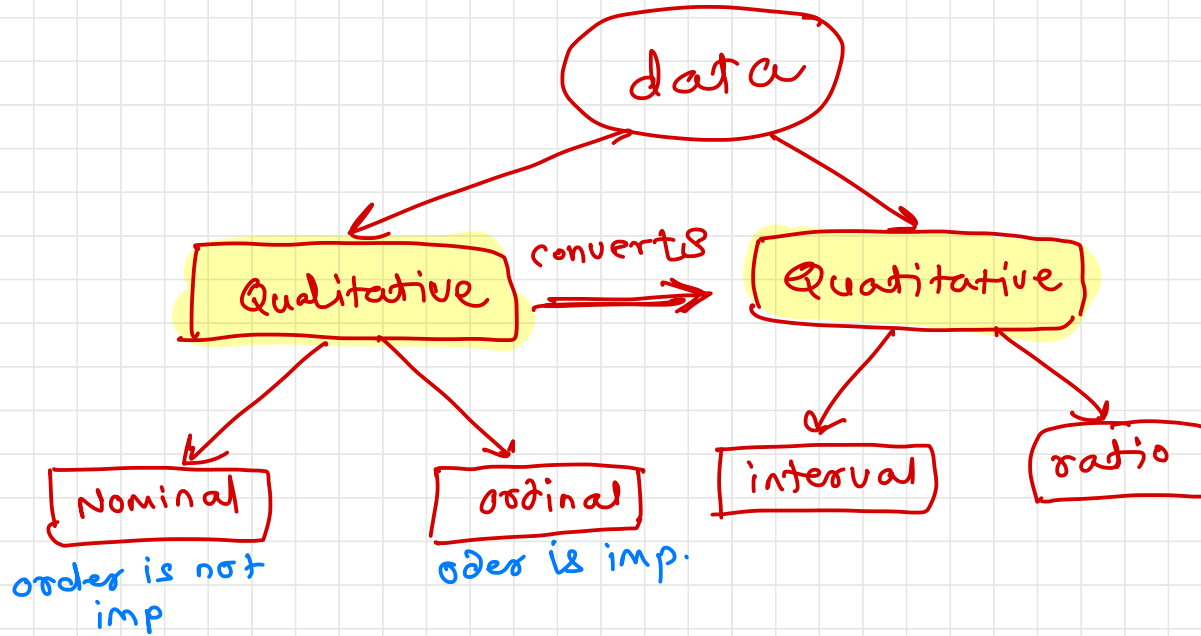
individual

variable

Name	Email	Phone	Address	Age
Person1	p1@test.com	7845343456	Pune	56
Person2	p2@test.com	6644664466	Mumbai	67
Person3	p3@test.com	7676767676	Satara	30
Person4	p4@test.com	8989898989	Karad	50

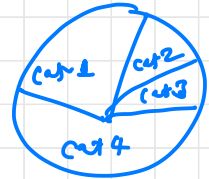
variable = Random Variable
= column
= attribute





Categoricals

cat1 = 100
cat2 = 200
cat3 = 150
cat4 = 200



categorical

city
pune
satara
pune
mun

Variables: Qualitative (categorical)

- Variables that are not measurement variables
- Also known as categorical variables [textual data]
- Take category or label values and place an individual into one of several groups
- Each observation can be placed in only one category ← mutually exclusive, →
→ colors
- The categories are mutually exclusive
- E.g. political party, profession, gender, whether person smokes

↓
20 20 40 30

observation = row

Types

- Nominal colors = [red, green, yellow, red, green]
 - Can not be ordered from smallest to largest
- Ordinal
 - Can be arranged in order of categories
 - But difference between data values can no be calculated or meaningless

worst not-so-good Good Fantastic ✓ excellent



Variables: Quantitative [numeric]

- Variables whose values result from counting or measuring something
- Also known as continuous variable
- Take numerical values and represent some kind of measurement
- E.g. height, weight, age

Types

- Interval
 - There is no true zero
- Ratio
 - There is a true zero

weight → 119
119.79, 119.78, 119.80,
120.04

Kelvin → true zero

celsius → not true zero

fahrenheit → not true zero.



Series

- Collection of data points is called as a series

- Types

- ✓✓ • Time Series

- A series of data that is arranged chronologically, or in relation to time

[second, minute, hour, day, week, year]



date	temp
12 apr 22	30°C
13 apr 22	32°C

- ✓✓ • Frequency Series

- A series of data that is formed along with the frequencies of their occurrences

- Types

- ✓ Individual series
- ✓ Discrete series
- ✓ Continuous series



Individual Series , [frequency : 1]

- Each value of the variable occurs for **only once**
- The frequency of occurrence of all the values in such a series is only one
- Such series are displayed **without the frequency column**
- E.g.
 - Marks: 40 60 ~~60~~ 80 45 85 90 67
 - Ages: 30 40 35 45 56 60

marks	optional frequency
40	1
60	1
80	1
45	1
85	1
90	1
67	1



Discrete series - [frequency $\neq 1$]

- The different values of a variable are shown in a discontinuous manner along with their respective frequencies
- Such a series can also be arranged either in ascending, or in descending order
- E.g.

frequency

Marks	# Students
80	2
40	4
45	3
60	6
86	3
90	3
70	10

frequency

Wages	# Workers
150	10
160	20
170	10
200	6
500	4
1000	2
1500	1

80, 80, 40, 40, 40, 40, 45, 45, 45, 60, 60, 60, 60, 60, 60, etc



Continuous Series

- Different values of the variables are stated in a continuous manner along with their respective frequencies
- Such series can be stated either in the form exclusive, or in the form of inclusive class intervals along with their respective class frequencies
- E.g.

10

Marks Range	# Students
10-20	5
20-30	0
30-40	20
40-50	15
50-60	10
60-70	15
70-80	10
80-90	20
90-100	5

Handwritten annotations: A red arrow points from the number '10' to the '20' in the '30-40' row. Another red arrow points from the '10' in the '10-20' row to the '20' in the '20-30' row. The '10-20' row has '10' crossed out and '19' written next to it. The '5' in the '# Students' column for the '10-20' row is circled.

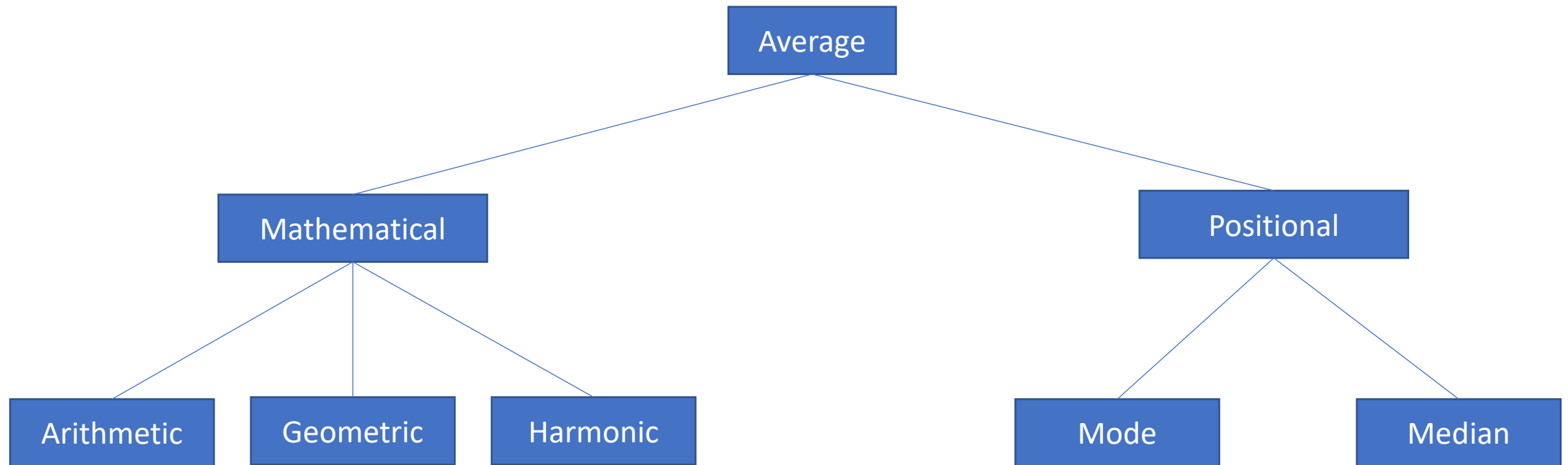


Measures of Central Tendency

- Is a summary statistic that represents the center point or typical value of a dataset
- These measures indicate where most values in a distribution fall and are also referred to as the central location of a distribution



Measures of central tendency



Arithmetic Average

- It is an average of the series
- Methods to calculate
 - Direct method
 - Assumed mean method
 - Step deviation method



Direct method

Individual Series

$$\bar{x} = \frac{\sum x}{n}$$

Discrete Series

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Continuous Series

$$\bar{x} = \frac{\sum fx}{\sum f}$$



Assumed mean method

- Assume any one as assumed mean
- Calculate deviation from assumed mean
- Find the sum of deviations from assumed mean
- Use the formula

Individual Series

$$\bar{x} = A + \frac{\sum(x - A)}{n}$$

Discrete Series

$$\bar{x} = A + \frac{\sum f(x - A)}{\sum f}$$

Continuous Series

$$\bar{x} = A + \frac{\sum f(x - A)}{\sum f}$$



Step deviation method

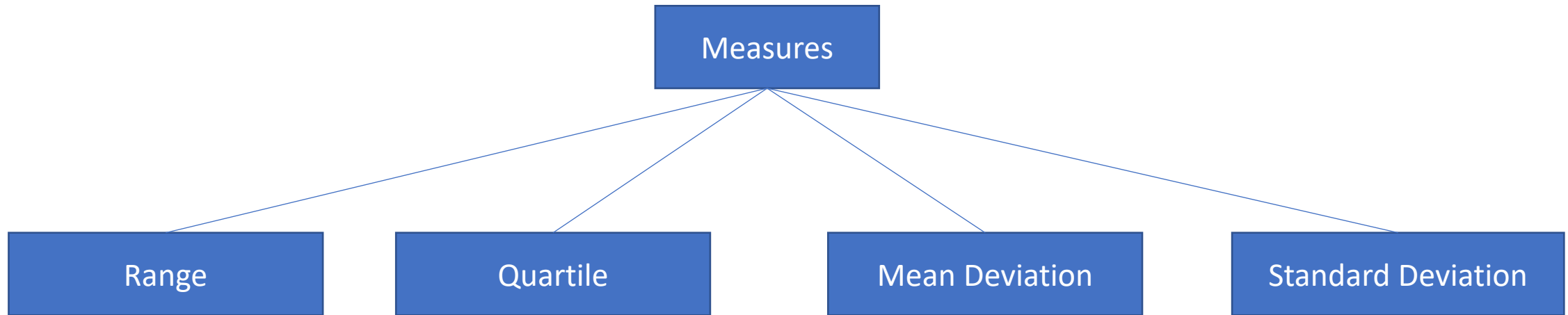
- Only possible in case of continuous series

$$\bar{x} = A + \frac{\sum df}{\sum f} * i$$



Measures of dispersion

- Measures dispersion in the data



Range

- The difference between largest and smallest value

$$\text{Range} = (L - S)$$



Partitioning

- Splitting the numbers in different partitions
- Can be done in following ways

$$k \frac{(n + 1)}{4}$$

Quantile

$$k \frac{(n + 1)}{10}$$

Decile

$$k \frac{(n + 1)}{100}$$

Percentile



Quartile Deviation

- Way to estimate the spread of a distribution about a measure of its central tendency (usually the mean)

$$\frac{Q3 - Q1}{2}$$



Mean Deviation

- Find mean of the series
- Find the absolute deviation
- Calculate $\sum(x - \text{mean})$

$$\frac{\sum(x - \bar{x})}{n}$$



Standard Deviation

- Used to quantify the amount of variation or dispersion of a set of data values
- A low standard deviation indicates that the data points tend to be close to the mean
- A high standard deviation indicates that the data points are spread out over a wider range of values

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

